

Lab 1: Software Success & Failure Analysis

Part A

PNG Case:

- **2022 PNG Election electoral roll- out-of-date rolls disrupted polling and disenfranchised voters.**

Part B

References

- Transparency International PNG INC. (2022, November 4). *TIPNG Dimestic Election Observation Report 2022*. Retrieved from <https://transparencypng.org.pg/cgi-sys/suspendedpage.cgi>
- Winn, T. (2022). *Papua New Guinea's elections: A review and some personal insights*. Griffith Asia Institute.

Part C

Question 1

What was the system meant to do, and for whom?

The parliamentary and electoral system that was used in Papua New Guinea was supposed to cater for eligible citizens with an unbiased, complimentary, inclusive and peaceful opportunity to vote for their representatives, ensure that the people voting are precisely counted, and produce a legitimate government that reflects the desire of the voters. The system was created to serve voters across all the provinces in the country including women, people with disabilities, rural communities, and other marginalized groups while promoting political stability, and effective representation in parliament. However, in the 2022 National Election, there were a lot of issues with the system which caused violence, delayed counting, there were many voters that were excluded, voters were intimidated to prevent them from voting, faulty electoral rolls, and administrative

failures. Therefore, this showed that the system failed to serve vulnerable voters which raised serious concerns about whether the election met the standards of being fair and free.

Question 2

What actually happened? Timeline of key events with your sources.

Timeline of events:

- Before the polling
 - There were political issues that caused poor electoral conduct in the 2022 elections. What happened was there was a delay in starting the Inter-Department Election Committee (IDEC), insufficient fundings and the late appointment of the Electoral Commissioner.
- Polling period
 - During the polling period there was a widespread number of problems that included double and multiple voting because TIPNG did not mark the voter's finger with indelible ink after they voted.
 - In the highlands, Bloc voting was taken as normal and was conducted by young men who collected up to 300 ballot papers and filled them, mainly people with disabilities, elderly people and women.
 - There were also other issues that were noticed during the polling period which includes electoral roll inaccuracies, underage voting, bribery, and intimidation.
- Counting and declaration period
 - An issue uncouned in this period was when William Powi, a contestant of the Southern Highlands Regional seat, was declared as the winning candidate even though allegedly not leading at the time counting was disrupted by violence which raised questions of the integrity of the election's outcome.
 - Another issue occurred in the Kabwum district in Morobe province where ballot boxes were burnt because the number of ballot papers did not

correspond to the total voting population which disrupted the counting process.

Question 3

Which of the four process activities broke down (or worked well): specification, development, validation, evolution? Justify.

➤ **Specification**

When looking into the specifications of this electoral system, the laws and regulations were specified clearly, but there was a disconnection between what happened in practice and what the law requires. This occurred because there were election laws that were breached such as electoral roll verification, voters' identification, ballot secrecy, and the election timeline was not properly followed. Therefore, the specifications broke down.

➤ **Development**

The development of the election system uncouneted a lot of issues with the implementations due to poor planning, delayed preparation, insufficient resourcing, late publication of schedules, and inaccurate electoral rolls. Therefore, the development process failed.

➤ **Validation**

The validation process had also failed because it was not what the people expected when it came to the election process. What the people expected was a smooth and improved system of election process unlike the previous elections but instead the 2022 elections were worse than the previous election. Hence, the election system was not valid, meaning it had failed also.

➤ **Evolution**

According to the report the same problems that were encountered in the previous elections and the recommendations that were stressed previously were not fully implemented showing a very poor evolution of the system.

Question 4

Which attributes of good software were violated or achieved: maintainability, dependability & security, efficiency, acceptability? Attributes (lecture slide 6)

The 2022 PNG national election electoral roll did not meet many of the attributes of a proper software programmed. Elector ability was prone because voter lists were no longer updated consistently during the election cycle, using election time followed incorrect and outdated voter data. Dependency and protection were additionally compromised because rolls could not be relied upon at all from which to discover qualified voters; Many legitimate citizens do not find their names on the list, at the same time reports suggest that some people may additionally be able to vote additional than once. Efficiency turned negative as roll practice was rapidly delayed before the election, adding additional administrative problems and confusion throughout voting. Finally, acceptability decreased because the machine does not serve the people it was designed for, mainly frustration, loss of confidence, and voter disenfranchisement. The indicators did not fully meet the standards of maintainability, dependability, efficiency, and acceptability.

Question 5

Name TWO software engineering practices that would most likely have changed the outcome and say why. Success factors (slides 8–9)

Two software engineering practices that could most probably have modified the final results of the 2022 PNG National Election electoral roll were continuous maintenance and making plans at some point of the electoral cycle, and daily validation and testing before deployment. The electoral roll modified into now not nicely maintained and become simplest finalized simply before the election, ensuing in defective voter records and lacking names. If the Electoral Commission had constantly saved voter information up to date and reviewed it at some stage in the five-12 months electoral cycle, masses of those errors could have been identified and corrected in advance. In addition, regular validation and trying out through audits, trial runs, and public verification of physical games could have helped locate replica facts, incorrect voter details, and lacking eligible citizens earlier than polling started. Together, those practices should have improved the accuracy and reliability of the electoral roll, reduced voter disenfranchisement, and elevated public self-warranty inside the election machine. They additionally cope with the crucial reasons of undertaking failure highlighted within the lecture, especially bad making plans, insufficient protection, and inadequate testing.

Question 6

PNG lens: what would an engineer in PNG learn from this case for local delivery conditions? Context-fit engineering (slides 10–11)

From a PNG angle, this example shows that successful software structures need to be designed to fit local conditions and supported by using strong statistics management tactics. Papua New Guinea's hard geography, faraway groups, confined infrastructure, and varying degrees of provincial potential mean that critical structures, which include the electoral roll, cannot depend on last-minute updates or centralized management on my own. Engineers in PNG should ensure that information is constantly maintained, demonstrated, and updated throughout the gadget's life cycle, as opposed to before important activities. The case also demonstrates that generating on my own isn't always sufficient; powerful planning, good enough funding, stakeholder involvement, and regular trying out are simply as essential. As highlighted in the lecture, the most successful systems in PNG aren't always the most advanced but the ones that are designed to work reliably within PNG's real-world constraints and people's wishes.